

BiteFixes Enterprise Architecture Manual

Volume 04

Domain Specifications

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Chapter 1

Domain Architecture

Definition

A Domain represents one business capability.

A Domain is **NOT**

- a table

- a REST endpoint
- a repository
- a Python package

A Domain represents a complete business responsibility.

Domain Structure

Every domain follows exactly the same architecture.

Domain

- ├ Purpose
- ├ Responsibilities
- ├ Entities
- ├ Value Objects
- ├ Aggregates
- ├ Use Cases
- ├ Policies
- ├ Events
- ├ Services
- ├ Repository Contracts
- ├ DTOs
- ├ APIs
- ├ Workflows

└─ Metrics

└─ Tests

This structure is mandatory.

Chapter 2

Company Domain

Purpose

Represents one independent business.

Responsibilities

The Company Domain owns:

- Company Profile
- Company Settings
- Branding
- Licensing
- Subscription
- AI Configuration
- Localization
- Working Hours
- Business Calendar
- Business Policies

Entities

Company

CompanySettings

CompanyBrand

CompanySubscription

CompanyLocalization

CompanyPolicy

Value Objects

Address

PhoneNumber

BusinessHours

Currency

Language

Timezone

TaxIdentifier

Aggregate Root

Company

Everything belongs to Company.

Use Cases

Create Company

Update Company

Activate Company

Suspend Company

Change Subscription

Configure Branding

Configure AI

Configure Localization

Business Rules

BR-001

Company must have one default language.

BR-002

Company must have one timezone.

BR-003

Company cannot be deleted.

Only archived.

Events

CompanyCreated

CompanyUpdated

SubscriptionChanged

BrandUpdated

LocalizationChanged

CompanySuspended

Metrics

Customers

Devices

Tickets

Revenue

AI Usage

Storage

Messages

Active Technicians

Chapter 3

Customer Domain

Purpose

Represents commercial relationships.

Responsibilities

- Customer Registration
- Customer Profile
- Preferences
- Communication
- Customer Status
- Customer History
- Customer Intelligence

Entities

Customer

CustomerProfile

CustomerPreference

CustomerHistory

CustomerScore

CustomerConsent

Value Objects

Email

Phone

Address

Language

ContactPreference

Aggregate Root

Customer

Use Cases

Register Customer

Update Profile

Deactivate Customer

Merge Customers

Block Customer

Search Customer

Calculate Customer Score

Business Rules

BR-101

Customer email must be unique inside the company.

BR-102

Blocked customers cannot create new tickets.

BR-103

Customer language defaults to company language.

Events

CustomerRegistered

CustomerUpdated

CustomerBlocked

CustomerMerged

CustomerArchived

Metrics

Lifetime Value

Customer Satisfaction

Average Resolution Time

Average Repair Cost

Open Tickets

Closed Tickets

Chapter 4

Device Domain

Purpose

Represents technological assets owned by customers.

Responsibilities

- Device Registration
- Diagnostics
- Technical Information
- Warranty
- Repair History
- Asset Lifecycle

Entities

Device

DeviceSpecification

Warranty

Diagnostic

RepairHistory

Accessory

Value Objects

IMEI

SerialNumber

MACAddress

OperatingSystem

DeviceCondition

Aggregate Root

Device

Use Cases

Register Device

Assign Owner

Start Diagnosis

Finish Diagnosis

Register Warranty

Archive Device

Business Rules

BR-201

Every device belongs to exactly one customer.

BR-202

Serial Number must be unique within a company.

BR-203

Diagnosis must exist before repair starts.

Events

DeviceRegistered

OwnerAssigned

DiagnosisStarted

DiagnosisCompleted

WarrantyActivated

RepairHistoryUpdated

Metrics

Repair Count

Average Repair Cost

Warranty Claims

Failure Rate

Brand Statistics

Model Statistics

Chapter 5

Domain Isolation

Each domain is autonomous.

Example:

Ticket Domain

↓

Customer Interface

↓

Customer Domain

Never:

Ticket

↓

Customer Database

↓

SQL Query

Domains communicate through contracts.

Chapter 6

Repository Contracts

Repositories expose business operations, not SQL.

Example:

CustomerRepository

↓

FindByUUID()

FindByPhone()

FindByEmail()

Save()

Archive()

Not:

```
SELECT *
```

```
FROM customers
```

SQL belongs to the Infrastructure layer.

Chapter 7

Domain Policies

Policies define business decisions.

Example:

```
VIP Customer Policy
```

↓

```
Repair Cost
```

↓

```
Greater than
```

↓

```
1000
```

↓

```
Manager Approval Required
```

Policies are configurable.

Chapter 8

Domain Services

Domain Services coordinate business logic that spans multiple entities.

Example:

EstimateService

↓

Ticket

↓

Device

↓

Inventory

↓

Labor Cost

↓

Generate Estimate

They never perform infrastructure work directly.

Chapter 9

Domain Integrity Rules

Every domain must guarantee:

- Consistency.
- Atomicity of business operations.
- Validation before persistence.
- Emission of domain events after successful execution.

No domain may leave the system in an inconsistent state.

Chapter 10

Domain Evolution Strategy

Each domain evolves independently.

Changes are categorized as:

- **Patch:** Internal implementation improvements.
- **Minor:** Backward-compatible business enhancements.
- **Major:** Business model changes requiring migration.

The goal is to evolve domains without forcing changes across the entire platform.

Enterprise Vision

The long-term objective is for every module of BiteFixes to be **replaceable without affecting the business model**.

That means:

- AI engines can change.
- Database technology can change.
- Mobile framework can change.
- Backend framework can change.

But the **domain specifications remain stable**.

They become the contract that defines what BiteFixes is, independent of how it is implemented.